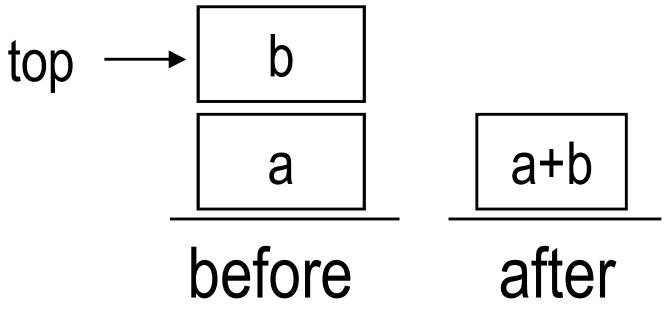
Stack effects

Informal description

OPERATION	STACK EFFECT	DESCRIPTION
e.g. +	(a b -- a+b)	add two topmost elements



Stack effect calculus – 1990-s

T - operand types (`char, flag, addr, ...`)

$\mathbf{T}^*$ - type lists (last type on the top)

Ø - type clash symbol (stack error)

The set of stack effects:

$$\mathbf{S} = (\mathbf{T}^* \times \mathbf{T}^*) \cup \{ \emptyset \}$$

$$(a \rightarrow b)$$

input parameters (types) output parameters (types)

Composition (multiplication)

For all s in **S**: $s \cdot \emptyset = \emptyset \cdot s = \emptyset$

For all a, b, c, d, e, f in $\mathbf{T}^*$:

$(a \to b) \cdot (eb \to d) = (ea \to d)$

$(a \to fc) \cdot (c \to d) = (a \to fd)$

$\emptyset$, otherwise

$\emptyset$ is zero

$1 = (\to)$ is unity for this operation

S is polycyclic monoid

Notation for rule based approach

t, u, … - types (just symbols)

$t \leq u$ – t is subtype of u (t is more exact) or equal to u (subtype relation is transitive)

$t \perp u$ - t and u are incompatible types

t^i - type symbols with "wildcard" index (index is unique for "the same type")

Notation (cont.)

a, b, c, d, … - type lists (top right) that represent the stack state

s = (a → b) – stack effect

(a – stack state before the operation, b – after)

Ø - type clash (zero effect)

Notation (cont.)

$(a \to b)\cdot(c \to d)$ - composition of stack effects $(a \to b)$ and $(c \to d)$ defined by rules

x, y – sequences of stack effects

y, where $u^j := t^k$ – substitution: all occurances of u^j in all type lists of sequence y are replaced by t^k, where k is unique index over y

Rules

$$\frac{x \cdot \varnothing}{\varnothing} \qquad \frac{\varnothing \cdot x}{\varnothing}$$

$$\frac{x \cdot (a \to b) \cdot (\ \to d)}{x \cdot (a \to bd)} \qquad \frac{x \cdot (a \to\ ) \cdot (c \to d)}{x \cdot (ca \to d)}$$

$$\frac{x \cdot (a \to bt) \cdot (cu \to d),\ \text{where}\ t \perp u}{\varnothing}$$

Rules (cont.)

$$\frac{x\cdot\left(a\rightarrow bt^{i}\right)\cdot\left(cu^{j}\rightarrow d\right),\ \text{where}\ t\leq u}{x\cdot\left(a\rightarrow b\right)\cdot\left(c\rightarrow d\right),\text{where}t^{i}:=t^{k}\ \ and\ u^{j}:=t^{k}}$$

$$\frac{x\cdot\left(a\rightarrow bt^{i}\right)\cdot\left(cu^{j}\rightarrow d\right),\ \text{where}\ u\leq t}{x\cdot\left(a\rightarrow b\right)\cdot\left(c\rightarrow d\right),\text{where}t^{i}:=u^{k}\ \ and\ u^{j}:=u^{k}}$$

Greatest lower bound

$$\frac{s \sqcap 0}{0} \qquad \frac{r \sqcap s}{s \sqcap r}$$

If there exist type lists $a_1, a_2, a_3, b_1, b_2, b_3, c_1, c_2, c_3$ such that for all elements of the lists these subtyping relations hold elementwise

$a_3 = min(a_1, a_2)$

$b_3 = min(b_1, b_2)$

$c_3 = min(c_1, c_2)$

then the following rule is applicable, in all other cases the result is zero.

$$\frac{(c_1 a_1 \to c_2 b_1) \sqcap (a_2 \to b_2)}{(c_3 a_3 \to c_3 b_3)}$$

Loop invariant

$$\sqcap^* s = s \sqcap (s \cdot s)$$

The result of this operation is an idempotent element that most precisely describes the loop body s.

Handling branches and loops

Greatest lower bound operation and loop invariants in use:

$$\frac{s(\ \texttt{IF}\ \alpha\ \texttt{ELSE}\ \beta\ \texttt{THEN}\ )}{(\texttt{flag} \rightarrow) \cdot [s(\alpha) \sqcap s(\beta)]}$$

$$\frac{s(\ \texttt{BEGIN}\ \alpha\ \texttt{WHILE}\ \beta\ \texttt{REPEAT}\ )}{\sqcap^* [s(\alpha) \cdot (\texttt{flag} \rightarrow)] \cdot \sqcap^* s(\beta)}$$

Example (small subset)

- Type system:

  a-addr < c-addr < addr < x

  flag < x

  char < n < x

Example (cont.)

- Words and specifications:

```
DUP    ( x[1] -- x[1] x[1] )
DROP   ( x -- )
SWAP   ( x[2] x[1] -- x[1] x[2] )
ROT    ( x[3] x[2] x[1] -- x[2] x[1] x[3] )
OVER   ( x[2] x[1] -- x[2] x[1] x[2] )
PLUS   ( x[1] x[1] -- x[1] )   "same type"
+      ( x  x  -- x )
@      ( a-addr  -- x )
!      ( x  a-addr  -- )
C@     ( c-addr  -- char )
C!     ( char  c-addr  -- )
DP     ( --  a-addr )
0=     ( n  -- flag )
```

Examples with control structures

```
: test1
   IF
      ROT
   ELSE
      @
   THEN ;

(a-addr[1] a-addr[1] a-addr[1] --
 a-addr[1] a-addr[1] a-addr[1])
```

Examples (cont.)

```
: test2
   BEGIN
      SWAP OVER
   WHILE
      NOT
   REPEAT ;
                  (flag[1] flag[1] -- flag[1] flag[1])
: test3
      OR FALSE SWAP ;
```